$$= \left(z_1\overline{z_1}\right)\left(z_2\overline{z_2}\right) = |z_1|^2\,|z_2|^2 \qquad \text{(by commutativity } z_2\overline{z_1} = \overline{z_1}z_2\text{)}$$

Therefore, $|z_1z_2| = |z_1||z_2|$.

Note

It can be generalized by means of mathematical induction to any finite number of terms:

$$|z_1z_2z_3\cdots z_n| = |z_1||z_2||z_3|\cdots|z_n|$$

That is the modulus value of a product of complex numbers is equal to the product of the moduli of complex numbers.

Similarly we can prove the other properties of modulus of a complex number.

Example 2.9

If $z_1 = 3+4i$, $z_2 = 5-12i$, and $z_3 = 6+8i$, find $|z_1|, |z_2|, |z_3|, |z_1+z_2|, |z_2-z_3|$, and $|z_1+z_3|$.

Solution

Using the given values for z_1, z_2 and z_3 we get $|z_1| = |3+4i| = \sqrt{3^2+4^2} = 5$

$$|z_2| = |5-12i| = \sqrt{5^2+(-12)^2} = 13$$

$$|z_3| = |6+8i| = \sqrt{6^2+8^2} = 10$$

$$|z_1+z_2| = |(3+4i)+(5-12i)| = |8-8i| = \sqrt{128} = 8\sqrt{2}$$

$$|z_2-z_3| = |(5-12i)-(6+8i)| = |-1-20i| = \sqrt{401}$$

$$|z_1+z_3| = |(3+4i)+(6+8i)| = |9+12i| = \sqrt{225} = 15$$

Note that the triangle inequality is satisfied in all the cases.

$|z_1+z_3| = |z_1|+|z_3| = 15$ (why?)

Example 2.10

Find the following (i) $\left|\dfrac{2+i}{-1+2i}\right|$ (ii) $\left|\overline{(1+i)}(2+3i)(4i-3)\right|$ (iii) $\left|\dfrac{i(2+i)^3}{(1+i)^2}\right|$

Solution

(i) $$\left|\frac{2+i}{-1+2i}\right| = \frac{|2+i|}{|-1+2i|} = \frac{\sqrt{2^2+1^2}}{\sqrt{(-1)^2+2^2}} = 1. \qquad \left(\because \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|},\ z_2 \neq 0\right)$$

(ii) $$\left|\overline{(1+i)}(2+3i)(4i-3)\right| = \left|\overline{(1+i)}\right|\,|2+3i|\,|4i-3| \qquad (\because |z_1z_2z_3| = |z_1||z_2||z_3|)$$

$$= |1+i|\,|2+3i|\,|-3+4i| \qquad (\because |z| = |\overline{z}|)$$

$$= \left(\sqrt{1^2+1^2}\right)\left(\sqrt{2^2+3^2}\right)\left(\sqrt{(-3)^2+4^2}\right)$$

$$= \left(\sqrt{2}\right)\left(\sqrt{13}\right)\left(\sqrt{25}\right) = 5\sqrt{26}.$$

(iii) $$\left|\frac{i(2+i)^3}{(1+i)^2}\right| = \frac{|i|\,|(2+i)^3|}{|(1+i)^2|} = \frac{1|2+i|^3}{|1+i|^2} = \frac{\left(\sqrt{4+1}\right)^3}{\left(\sqrt{2}\right)^2} \quad \left(\because \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|},\ z_2 \neq 0\right)$$

$$= \frac{\left(\sqrt{5}\right)^3}{2} = \frac{5\sqrt{5}}{2}.$$

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